CodeLlama 4309: natural-text expression

50 regressions: original base output passes; original merged output fails. Joint sparse latent measured on each text.

Measurement	Base-generated text	Merged-generated text
Tasks with any activation	50/50	44/50
Mean task-mean activation	0.1997	0.1979
Median first activation (active texts only)	31.5%	58.9%
First activation after main-function end	0/50	0/50

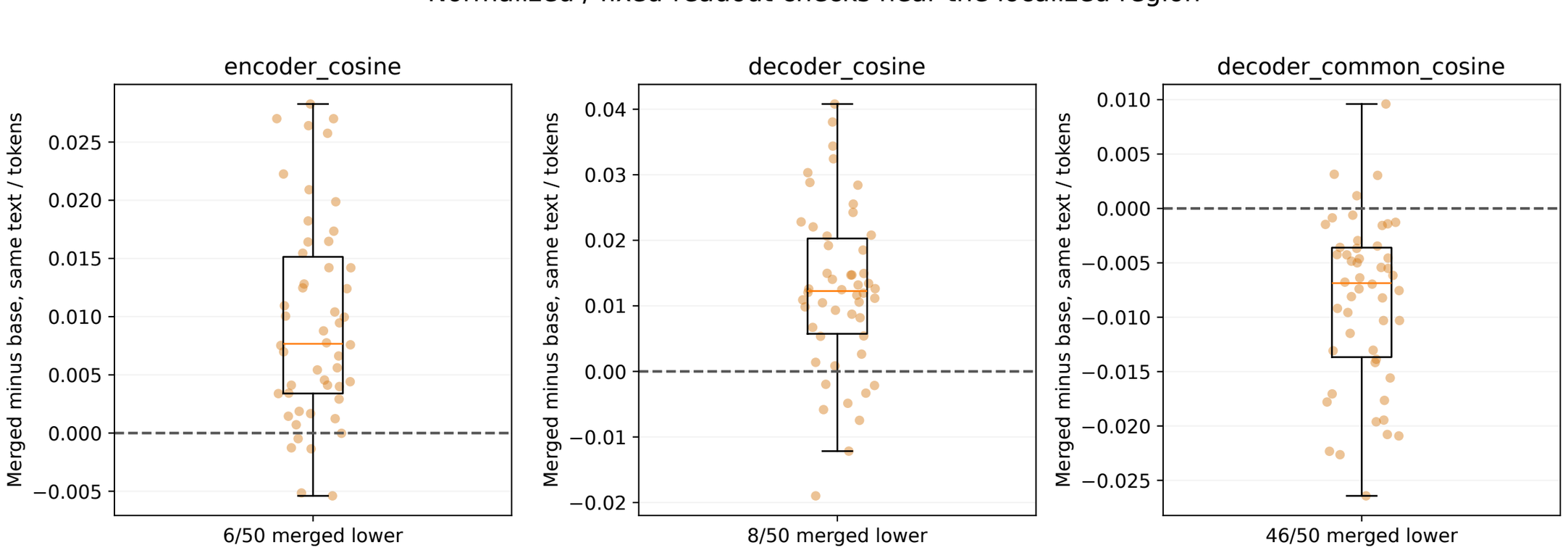
The shared latent is NOT generally stronger on failing merged texts by task-mean activation.
First-position percentages compare different generated programs and only active cases.
After-function excludes the newline ending the function itself; a return-line newline is not after-function.

Same text: base versus merged encoder contributions

Both models replay exactly the same original failing merged text. Contributions precede the JOINT ReLU / TopK.

Region	Base contribution	Merged contribution	Merged lower
Whole evaluated code	-0.999	-0.428	2/50
10 tokens before localized region start	-0.806	-0.211	12/50
Region start + next 10 tokens	-1.239	-0.591	5/50
Region start +/-10 tokens	-1.129	-0.504	4/50

Anchor is NOT the uniform DeepSeek test/import marker. It reuses existing failure localization:
16 runtime execution-site cases, 18 provisional line/interval cases and 16 omission regions.
48 anchors transfer through exact text blocks; /115 and /423 are manually localized on frozen text.
All 50 tasks enter local analysis. These are NOT independent sparse activations.


Normalized / fixed-readout checks near the localized region
encoder_cosine
decoder_cosine
decoder_common_cosine
Merged minus base, same text / tokens
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
6/50 merged lower
8/50 merged lower
46/50 merged lower
Own encoder / own decoder directions, then one common decoder direction for both models.
Geometric readouts are not calibrated independent feature activations; selected exploratory cohort, no mediation claim.